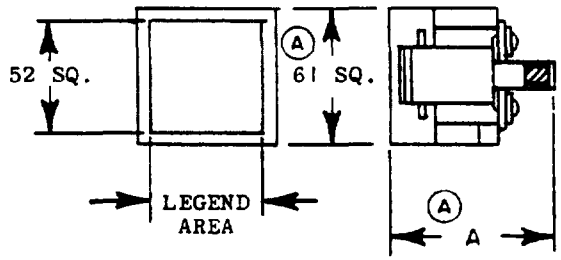
MILITARY SPECIFICATION SHEET

SWITCH, PUSHBUTTON, 4-LAMP, PUSHBUTTON AND LENS ASSEMBLIES FOR, .610 SQUARE (DRIPPROOF, SUNLIGHT READABLE, EMI/RFI SHIELDED)

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the pushbutton and lens assembly described herein shall consist of this specification and the latest issue of MIL-S-22885.

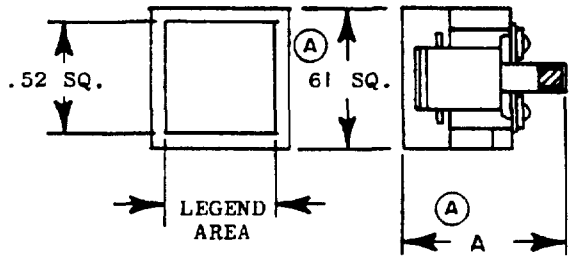
INCHES	MM
.52	13.21
.61	15.49
.71	18.03
1.01	25.65



LAMP CIRCUIT 1 and 4
Style A - Without EMI/RFI Shielding
Style E - With EMI/RFI Shielding

(A)

	DIM A
SHORT VERSION	.71
EXTENDED VERSION	1.01



LAMP CIRCUIT 3
Style B - Without EMI/RFI Shielding
Style F - With EMI/RFI Shielding

FIGURE 1. PUSHBUTTON STYLES, (UNSEALED) A, B, E, F

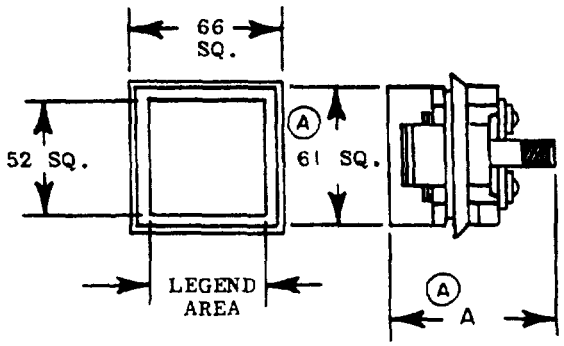
NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals
3. Lamps shown for reference only.
4. The design configuration is optional within the envelope dimensions shown.

(A) denotes changes

FSC 5930

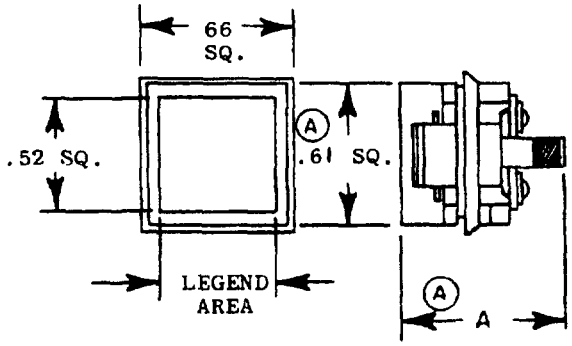
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:34Z
Check the source to verify that this is the current version before use.

INCHES	MM
.52	13.21
.61	15.49
.66	16.76
.71	18.03
1.01	25.65



LAMP CIRCUIT 1 and 4
Style C - Integral Seal
Style G - Integral Seal/EMI/RFI Shielding

		DIM A
A	SHORT VERSION	.71
	EXTENDED VERSION	1.01



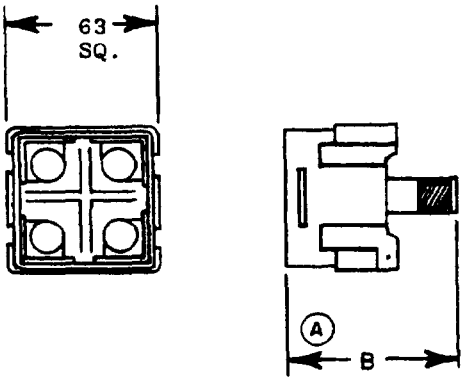
LAMP CIRCUIT 3
Style D - Integral Seal
Style H - Integral Seal/EMI/RFI Shielding

FIGURE 2. PUSHBUTTON STYLES, (SEALED) C, D, G, H

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ± 010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
3. Lamps shown for reference only.
4. The design configuration is optional within the envelope dimensions shown.

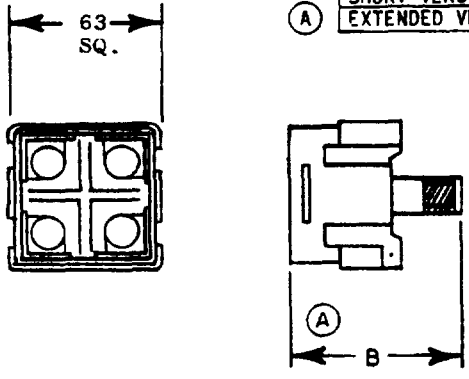
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:34Z
Check the source to verify that this is the current version before use.

INCHES	MM
63	16 00
69	17.53
99	25 14



LAMP CIRCUIT 1 and 4
Style A - Without EMI/RFI Shielding
Style E - With EMI/RFI Shielding 4/

	DIM B
SHORT VERSION	.69
EXTENDED VERSION	.99

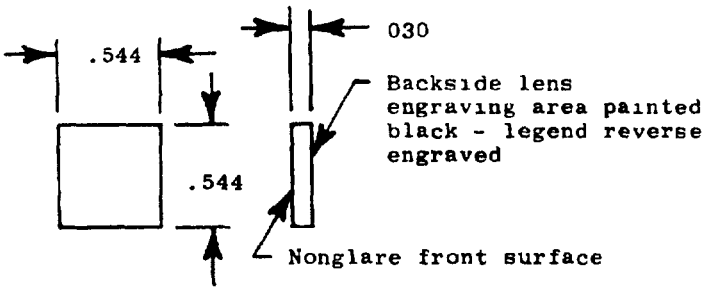


LAMP CIRCUIT 3
Style B - Without EMI/RFI Shielding
Style F - With EMI/RFI Shielding 4/

FIGURE 3. SPARE PART, PUSHBUTTON STYLES, (UNSEALED) A, B, E, F

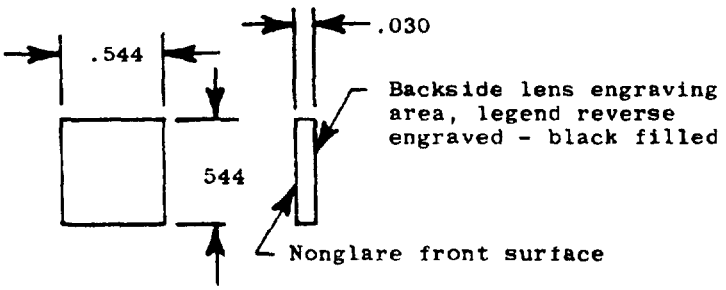
NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
3. The design configuration is optional within the envelope dimensions shown.
4. EMI/RFI screen not shown for clarity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:34Z
Check the source to verify that this is the current version before use.

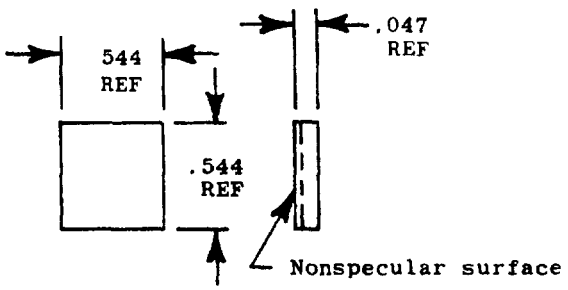


INCHES	MM
030	762
.047	1 194
544	13.818

LENS STYLE 1: FRONT LENS PLATE, TYPE N



LENS STYLE 2: FRONT LENS PLATE, TYPE W AND C



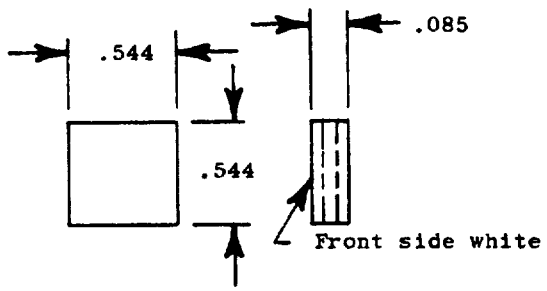
LENS STYLE 3: FRONT LENS PLATE, LAMINATED, TYPE S 3/

FIGURE 4. SPARE PART, FRONT LENS PLATE

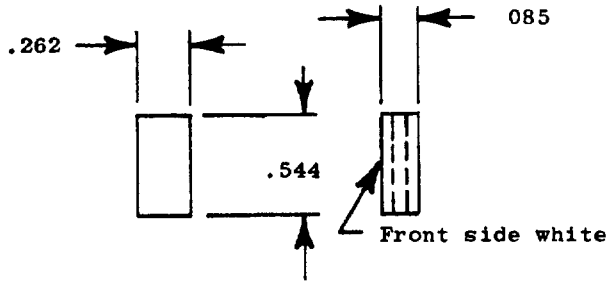
NOTES:
1. Dimensions are in inches
2. Unless otherwise specified, tolerances are ± .010 for three place decimals and ± 03 for two place decimals
3/ For reference only Lens Style 3 must be procured as total laminated assembly of lens Type S.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:34Z
Check the source to verify that this is the current version before use.

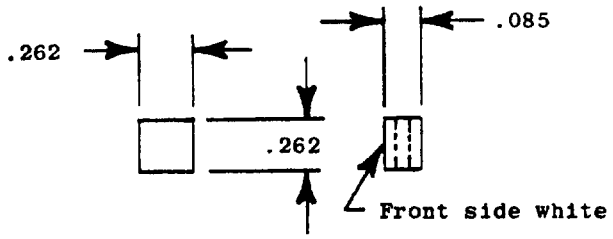
INCHES	MM
.085	2 159
.262	6.655
.544	13.818



LENS STYLE 4· LAMINATED FILTER



LENS STYLE 5: LAMINATED FILTER

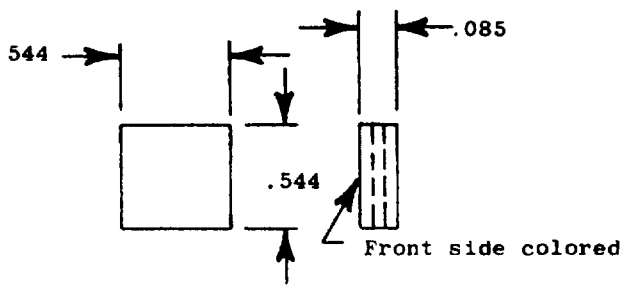


LENS STYLE 6· LAMINATED FILTER

FIGURE 5. SPARE PART, LAMINATED FILTER, TYPE N AND TYPE W

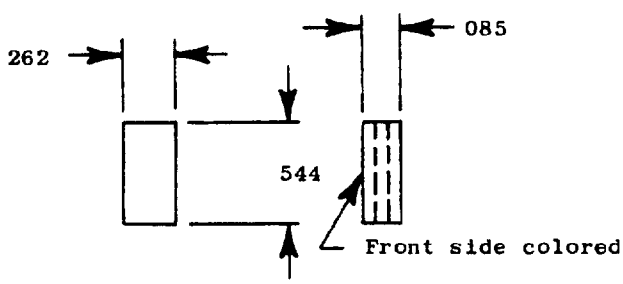
NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ± .010 for three place decimals and ± .03 for two place decimals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:34Z
Check the source to verify that this is the current version before use.

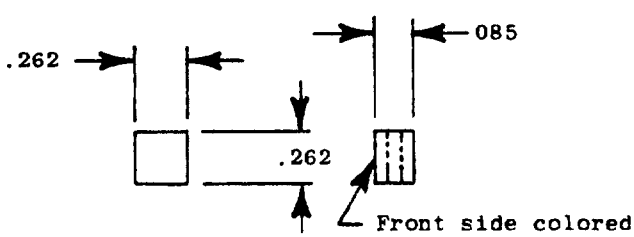


INCHES	MM
085	2 159
.262	6.655
544	13.818

LENS STYLE 7: LAMINATED FILTER



LENS STYLE 8· LAMINATED FILTER



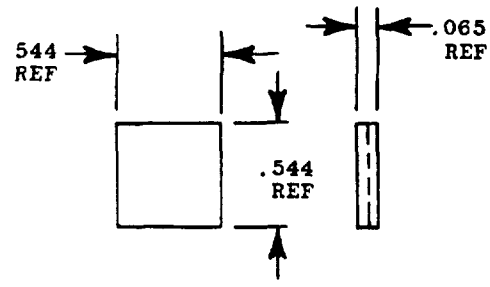
LENS STYLE 9: LAMINATED FILTER

FIGURE 6. SPARE PART, LAMINATED FILTER, TYPE C

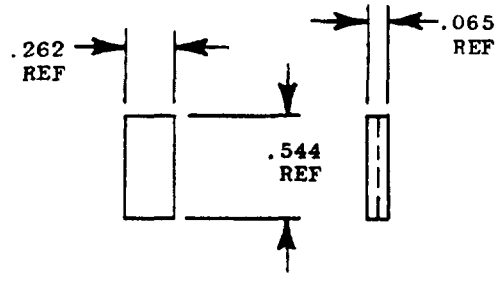
NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ± .010 for three place decimals and ± .03 for two place decimals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:34Z
Check the source to verify that this is the current version before use.

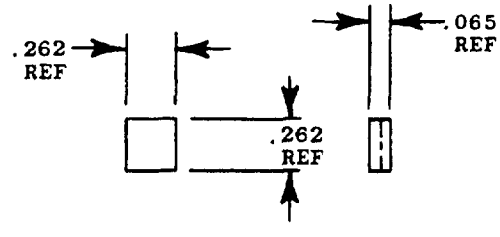
INCHES	MM
.065	1.651
262	6.655
544	13.818



LENS STYLE 10· LAMINATED FILTER



LENS STYLE 11 LAMINATED FILTER

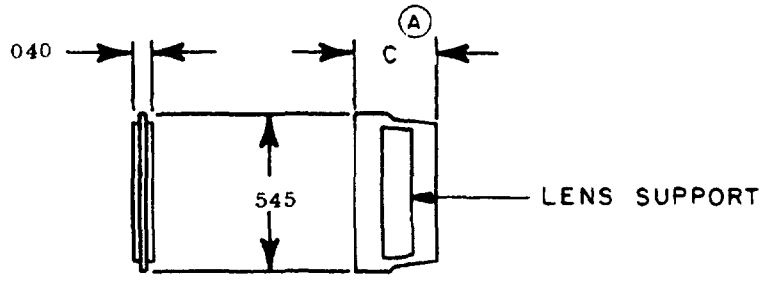


LENS STYLE 12: LAMINATED FILTER

FIGURE 7. SPARE PART, LAMINATED FILTER, TYPE S 3/

NOTES·
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ± .010 for three place decimals and ± .03 for two place decimals
3/ For reference only. Lens Style 10, 11, and 12 must be procured as total laminated assembly of lens Type S.

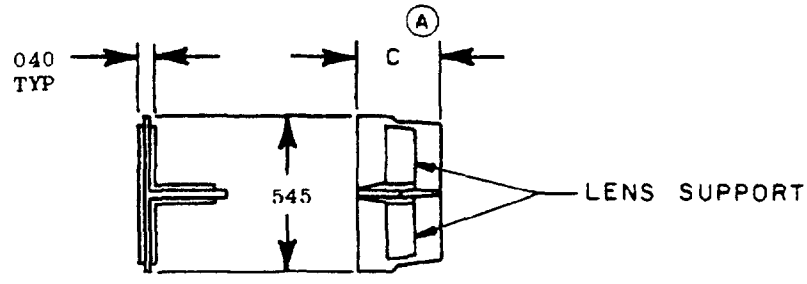
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:34Z
Check the source to verify that this is the current version before use.



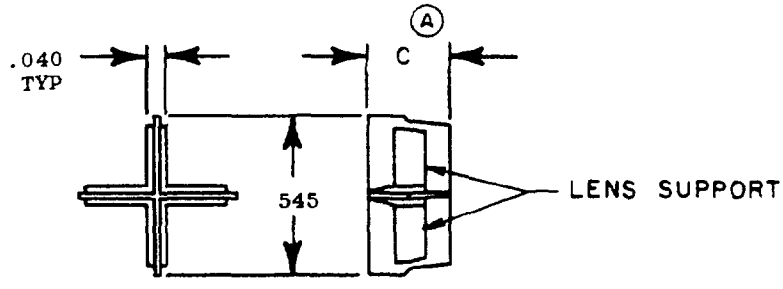
INCHES	MM
.040	1.02
.252	5.40
.545	13.84
.552	14.02

2-WAY DIVIDER

		DIM C
(A)	SHORT VERSION	.252
	EXTENDED VERSION	.552



3-WAY DIVIDER

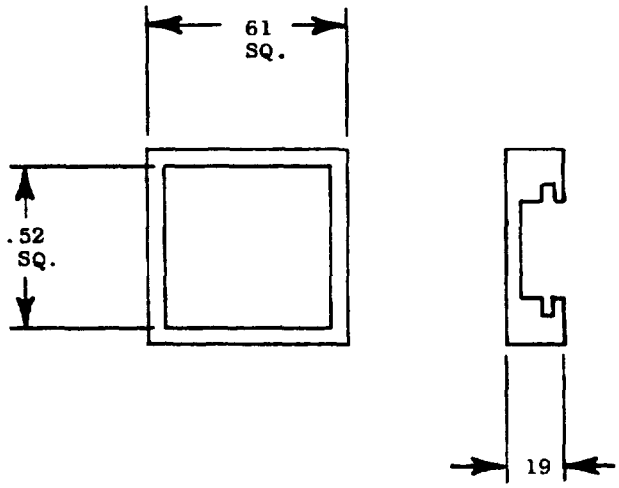


4-WAY DIVIDER

NOTES
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
3. Type S dividers cannot be ordered separately.
(A) 4. Design of lens supports optional within envelope dimensions shown.

FIGURE 8. Spare part, divider, type W, N and C. (See note 3)

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INCHES	MM
.19	4.83
52	13 21
61	15 49

FIGURE 9. SPARE PART, LENS RETAINER

NOTES:
1. Dimensions are in inches
2 Unless otherwise specified, tolerances are ± 010 for three place decimals and ± .03 for two place decimals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:34Z
Check the source to verify that this is the current version before use.

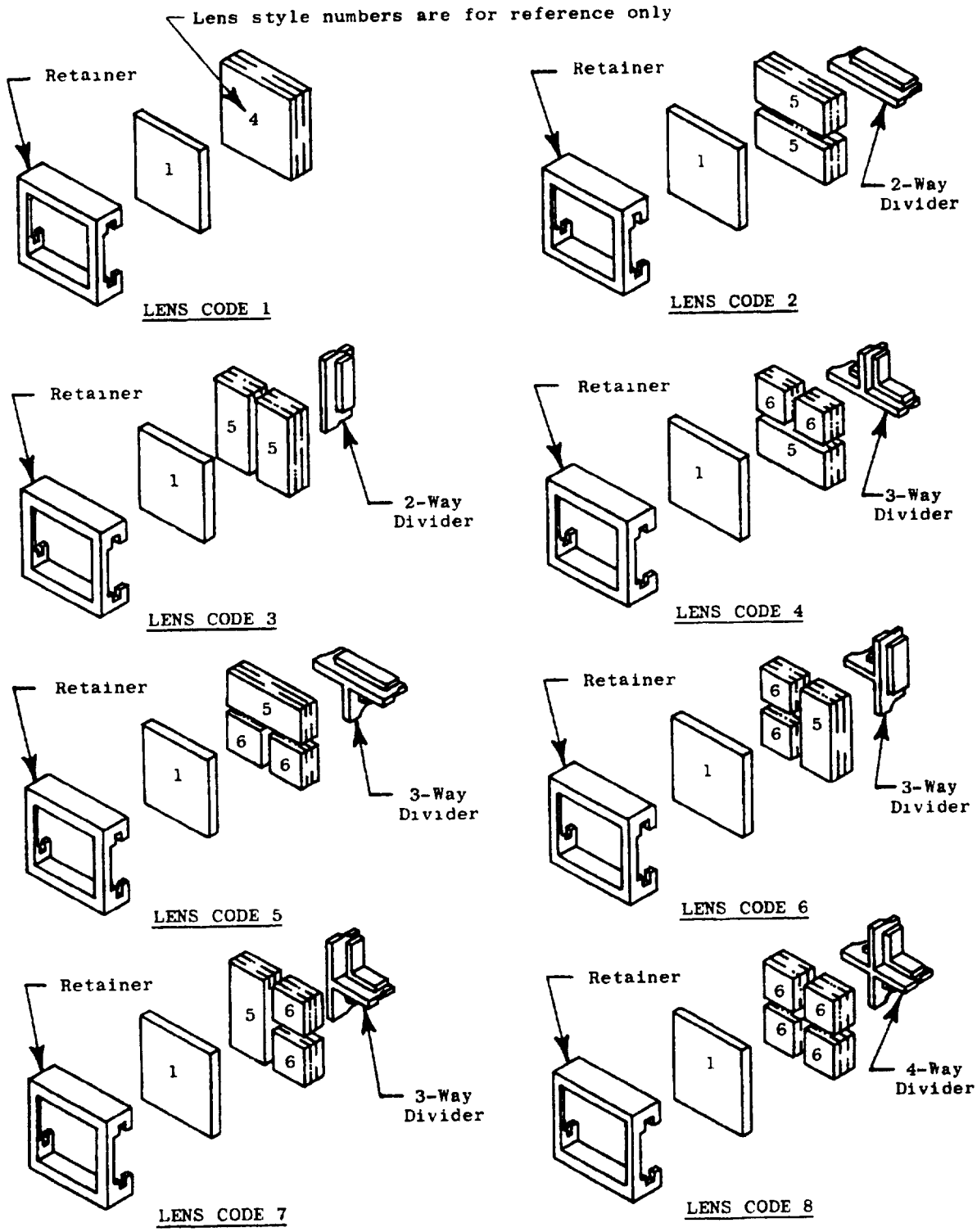


FIGURE 10. LENS ASSEMBLY DIAGRAM, TYPE N

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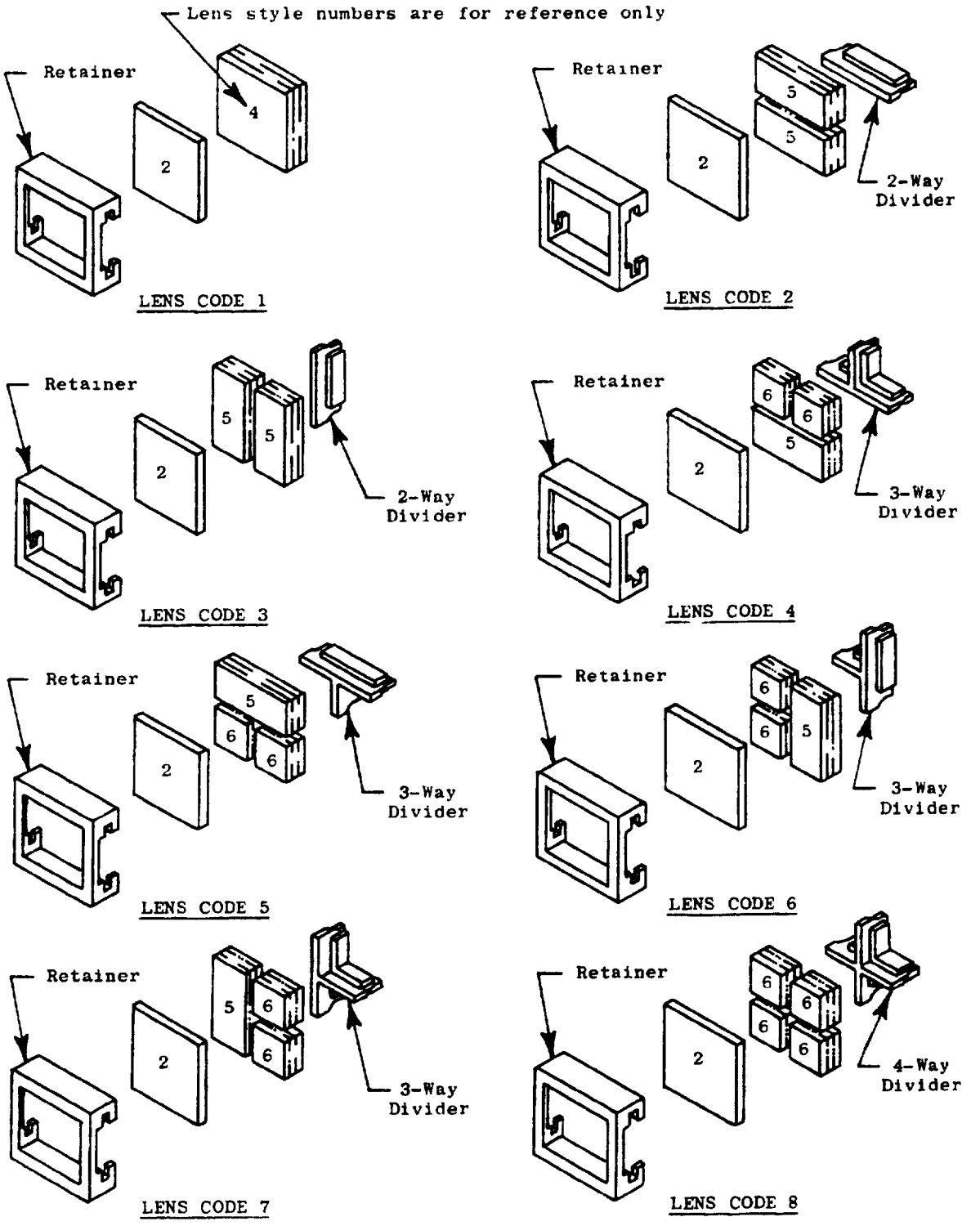


FIGURE 11 LENS ASSEMBLY DIAGRAM, TYPE W

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:34Z
Check the source to verify that this is the current version before use.

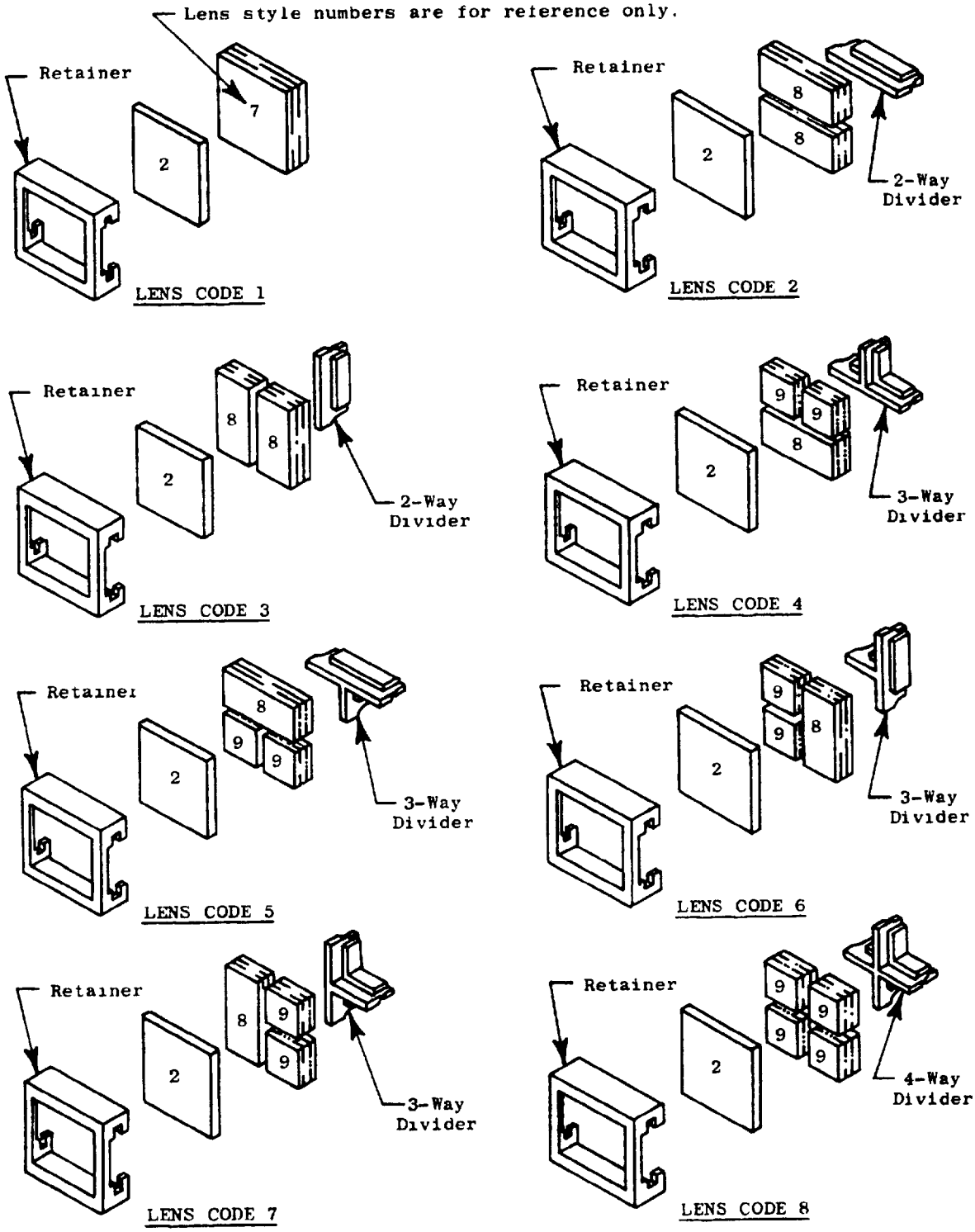


FIGURE 12. LENS ASSEMBLY DIAGRAM, TYPE C

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:34Z
Check the source to verify that this is the current version before use.

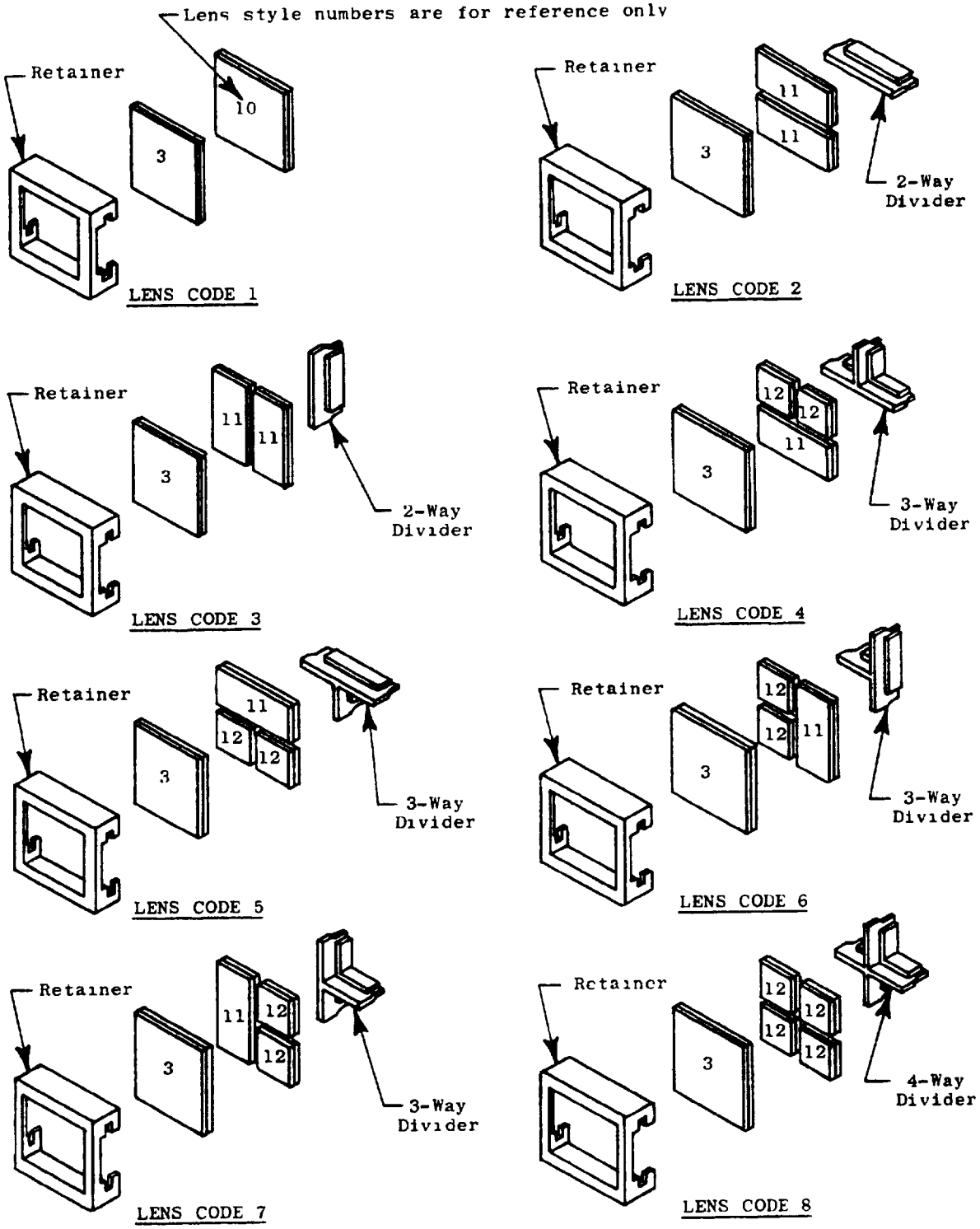


FIGURE 13 LENS ASSEMBLY DIAGRAM, TYPE S 1/

1/ Lens assembly diagram for Type S displays for reference only Type S displays shall be procured as complete laminated assemblies

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Ⓐ TABLE I. Illuminated color limits. 3/

Color	X 1/	Y 1/
Red (R)	.665 .663 .711 .713	SL 2/ .335 .287 SL 2/
Green (G)	.300 .300 .365 .365	.560 SL 2/ .560 SL 2/
Yellow (Y)	.568 .575 .630 .623	.425 SL 2/ SL 2/ .370
Blue (B) 4/	.250 .250 .330 .330	.330 .420 .330 420
White (W)	.480 .480 .540 .540	.395 .435 .431 .391

1/ The chromaticities of the colors are expressed as "X" and "Y" coordinates on the CIE chromaticity diagram and are within the limits bound by the coordinates listed for each color.

2/ Where intersection occurs with the spectrum locus on the CIE chromaticity diagram.

3/ Chromaticity values are obtained when switch is illuminated by 4 (T-1 midget-flange base) subminiature lamps of .15 ±.01 mean spherical candlepower (MSCP) at 5 V dc, 2400°K

4/ Not available on type S with EMI/RFI screen.

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Check the source to verify that this is the current version before use.

REQUIREMENTS

Intended use As pushbutton and lens assemblies for 4-lamp illuminated pushbutton switches, MIL-S-22885/100 and MIL-S-22885/101.

Dimensions and configurations See figures 1 through 13.

Pushbutton

Complete pushbutton shall consist of:

1 - Pushbutton Includes integral seal and EMI/RFI screen when specified.

1 - Front lens and color filter assembly, lens retainer and divider, as specified.

Ⓐ 4 - Lamps (T-1 submidget-flange base) not included, order separately. Industry Lamp No. 6180 is recommended (5 volt, .060 amps, .15 MSCP). Other lamps are available with various power requirements and illumination results. When using higher wattage lamps, additional heat sinking and air flow must be provided (use extended version of switch), and matrix mounting is not recommended.

Lens assemblies

Code 1 consists of

1 - Retainer
1 - Front lens plate
1 - Laminated filter

Code 2 and 3 consist of

1 - Retainer
1 - Front lens plate
2 - Laminated filters
1 - Divider, 2-way split

Code 4, 5, 6, and 7 consist of

1 - Retainer
1 - Front lens plate
3 - Laminated filters
1 - Divider, 3-way split

Code 8 consists of

1 - Retainer
1 - Front lens plate
4 - Laminated filters
1 - Divider, 4-way split

Materials and finishes.

Ⓐ Lens retainer Corrosion-resistant steel, black paint.

Pushbutton and divider High temperature thermoplastic.

Lamp buss Copper base alloy - gold plated.

Front lens plate and laminated filter High temperature heat resistant plastic. Thermal shock test is performed in accordance with MIL-S-22885, the maximum temperature shall be +85°C.

Characteristics See table IV.

Color See table I For brightness and chromaticity values, see MIL-S-2885/100 and /101

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:34Z
Check the source to verify that this is the current version before use.

Qualification inspection Required as part of the 4-lamp illuminated pushbutton switch, M22885/100 or /101

Group A inspection Shall be in accordance with MIL-S-22885, visual and mechanical only.

Ⓐ Part number

Short version M22885/102- (dash number from table II).

Extended version M22885/102E (dash number from table II).

For acquisition of government spares, M22885/102 or M22885/102E (dash numbers from tables II and III) shall be used

Ⓐ Part numbers Part numbers for complete pushbutton are assigned as follows:

Extended version	M22885/102E	25	S	2GR
Short version	M22885/102-	25	S	2GR
	Specification number	Pushbutton config (see table II)	Display type (see table IV) 2/	Display config. (see table V)
	Standard pushbutton, no seal, no EMI/RFI shielding, Lamp Ckt. 1 and 4, Display Type S		Sunlight readable Display	Display code 2. Horiz. split, green color filter top half, red color filter bottom half

These part numbers are for the convenience of users only and do not include legends. Acquire legend portion of assembly separately from QPL source. For acquisition of government spares, use dash numbers shown in tables II and III.

TABLE II. Pushbutton configuration dash numbers.

Display type					Pushbutton style	Lamp 4/ circuit
N	W	C	S	1/		
Standard pushbutton, no seal, no EMI/RFI shielding						
01	09	17	25	33	A	1 and 4
02	10	18	26	34	B	3
Standard pushbutton with seal, no EMI/RFI shielding						
03	11	19	27	3/	C	1 and 4
04	12	20	28		D	3
Standard pushbutton with EMI/RFI shielding, no seal						
05	13	21	29	35	E	1 and 4
06	14	22	30	36	F	3
Standard pushbutton with seal and EMI/RFI shielding						
07	15	23	31	3/	G	1 and 4
08	16	24	32		H	3

1/ For pushbutton spare parts, order 33 through 36, as required. Does not include lens retainer, lens assemblies, or dividers.

2/ Reference letter code for display type. Must be used in conjunction with appropriate dash numbers, i.e., -25S, -01N, -09W, etc.

3/ Spare pushbutton styles C, D, G and H not available.

4/ Lamps circuits per MIL-S-22885/100 and /101, for reference only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Spare part configuration part number.

Part no. M22885/102-	Display type	Lens 3/ code	Style 4/ number	Color	Description
37	N	All	1	Clear	Front lens plate Painted black
38	W, C	All	2	Clear	Front lens plate
39	N, W	1	4	White	Filter, full display
40	"	"	"	Red	"
41	"	"	"	Green	"
42	"	"	"	Yellow	"
43	"	"	"	Blue	"
44	"	2,3,4,5,6,7	5	White	Filter, half display
45	"	"	"	Red	"
46	"	"	"	Green	"
47	"	"	"	Yellow	"
48	"	"	"	Blue	"
49	"	4,5,6,7,8	6	White	Filter, qtr. display
50	"	"	"	Red	"
51	"	"	"	Green	"
52	"	"	"	Yellow	"
53	"	"	"	Blue	"
54	C	1	7	White	Filter, full display
55	"	"	"	Red	"
56	"	"	"	Green	"
57	"	"	"	Yellow	"
58	"	"	"	Blue	"
59	"	2,3,4,5,6,7	8	White	Filter, half display
60	"	"	"	Red	"
61	"	"	"	Green	"
62	"	"	"	Yellow	"
63	"	"	"	Blue	"

See footnotes at end of table.

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Check the source to verify that this is the current version before use.

TABLE III. Spare part configuration part number - Continued.

Part no. M22885/102-	Display type	Lens 3/ code	Style 4/ number	Color	Description
64	C	4,5,6,7,8	9	White	Filter, qtr. display
65	"	"	"	Red	"
66	"	"	"	Green	"
67	"	"	"	Yellow	"
68	"	"	"	Blue	"
69 5/	All	2, 3	---	---	Divider, 2-way
70 5/	All	4,5,6,7	---	---	Divider, 3-way
71 5/	All	8	---	---	Divider, 4-way
72	All	All	---	---	Lens retainer
73	S	All	All	All	Lens assembly
1/	S	All	3 2/	---	Front lens plate
1/	S	1	10 2/	All	Filter, full display
1/	S	2,3,4,5,6,7	11 2/	All	Filter, half display
1/	S	4,5,6,7,8	12 2/	All	Filter, qtr. display

1/ Type S displays shall be procured as complete assemblies only, that consist of front lens plate, filters, and dividers. To order, specify. M2885/102-73S or M22885/102E73S (plus display configuration and color(s)) as required.

2/ Lens style 3, 10, 11, and 12 are shown for reference only.

3/ See table V and figures 10, 11, 12 and 13.

4/ See figures 4, 5, 6 and 7.

5/ When ordering extended dividers, specify M22885/102E (dash number 69, 70, or 71, as required).

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TABLE IV Display type.

Display type	Description
N	LIGHTED LETTERS Letters appear white on a black background until illuminated and then letters appear in color, background remains black.
W	LIGHTED BACKGROUND Letters appear black on a white background until illuminated and then background appears in color, letters remain black
S	SUNLIGHT READABLE Letters are not legible until illuminated and then letters appear in color, background remains black. When illuminated, lighted letters are readable in direct sunlight.
C	COLORED BACKGROUND Letters appear black against a colored background until illuminated and then background appears in lighted color, letters remain black

TABLE V. Display configuration. 2/

Display code 1/ 3/	Display
1X	
2XX	
3XX	
4XXX	
5XXX	
6XXX	
7XXX	
8XXXX	

1/ Display codes shown are for pushbuttons without color filters Replace X's with the color symbols of table I in sequence shown.
2/ Legends shall be procured separately from a source listed on the qualified products list.
3/ Display codes 3XX through 8XXXX not available with pushbutton styles B, D, F and H.

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Custodian
Air Force - 85

Review activities
Air Force - 11, 99

User activity:
Air Force - 19

Agent
DLA - ES

Preparing activity:
Air Force - 85

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